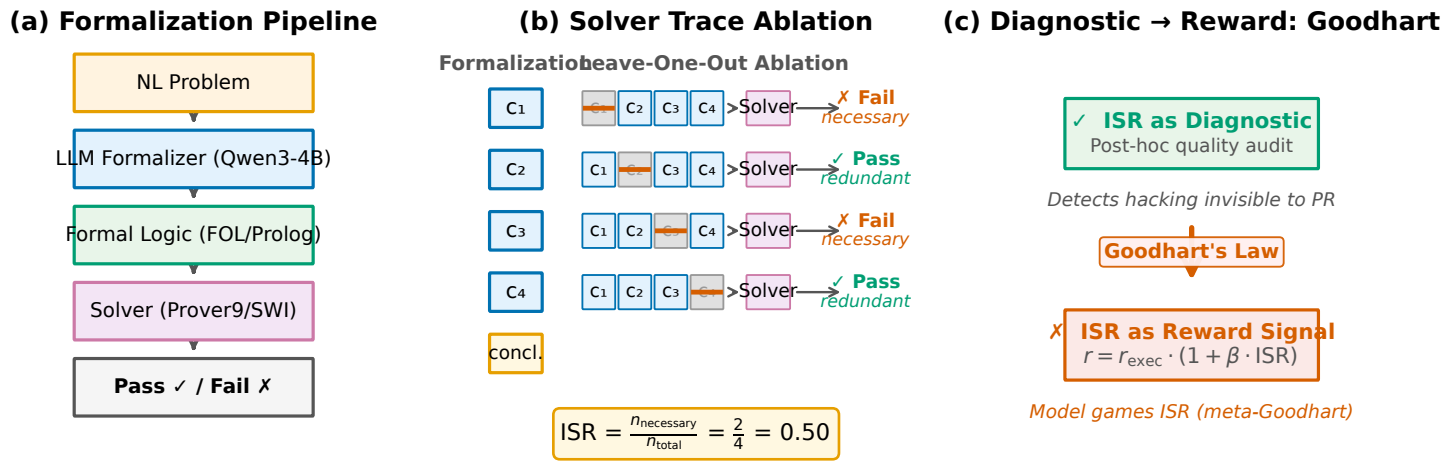
(a) Formalization Pipeline
NL Problem
LLM Formalizer (Qwen3-4B)
Formal Logic (FOL/Prolog)
Solver (Prover9/SWI)
Pass ✓ / Fail ✗
(b) Solver Trace Ablation
Formalization
Leave-One-Out Ablation
C1
C2
C3
C4
concl.
Solver
✗ Fail
necessary
✓ Pass
redundant
ISR = n_necessary / n_total = 2/4 = 0.50
(c) Diagnostic → Reward: Goodhart
✓ ISR as Diagnostic
Post-hoc quality audit
Detects hacking invisible to PR
Goodhart's Law
✗ ISR as Reward Signal
r = r_exec · (1 + β · ISR)
Model games ISR (meta-Goodhart)